

Medication Management in the Making: On Ethnography-Design Relations

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ABSTRACT

The relation between ethnography and design is often discussed in terms of being direct or indirect. The debate on using ethnography in design, models the problem as a matter of mediating between users and designers. This fails to take ‘use’ serious and renders the problem epistemic i.e. a matter of creating a better understanding or description of the user. Inspired by later developments in Science and Technology Studies I engage an ontological reconceptualization and turn to consider and practice the relation as performative – thus making ethnography, design and users’ practices converge. I show this by the case of how the concept of ‘medication management’ has been performed differently on a combined CSCW and participatory design project in healthcare. It is suggested that through design interventions with working prototypes; prospective analysis and participatory design can be fruitfully assembled in situations of use.

Author Keywords

Ethnography; participatory design; design intervention; design-in-use; prospective analysis; performative ontology; ontological convergence; medication management.

ACM Classification Keywords

D.2.2; H.5.3; J.3; J.4

INTRODUCTION

CSCW may be viewed as a response to the challenge of designing ‘better’ or more useful computer systems. Sociologists and anthropologist, in particular, have radically reformulated this challenge as being a matter of better understanding the broader social and organizational context of cooperate activities. However, in Schmidt and Bannon’s [38] programmatic call for CSCW they advocate a reorientation of social science and emphasize that if CSCW is to be taken seriously, CSCW research should not be descriptive but “constructive”. They suggest investigating exactly how insights from studies of cooperative work might be applied in the design of useful computer systems: “[T]he objective of social science contributions to CSCW

should not be to cash in on the new wave and do what they have always done but rather to explore exactly how insights springing from studies of cooperative work relations might be applied and exploited in the design of useful CSCW systems.” [ibid.]

A response to this call can be found in the long-standing debate on the role of ethnography in systems design. Some argue for the analytic purchase of ethnography while others are more critical and argue that “*designers do not need ethnography to do what they wish to do*” [5]. Either way, the logic of the debate models the problem as being about user-designer relations. This, I argue, is unproductive because it essentially fails to take an interest in use and because it renders the ethnography-design relation to be a matter of creating representations (such as analytic descriptions).

In this paper I draw on Science and Technology Studies (STS) to engage an *ontological reconceptualization* [25] of the problem of how to mobilize ethnography in design. I suggest working towards what could be termed ‘ontological convergence’ by creating situations where ethnography, design and users’ practices can be performed as mutually constitutive. By a short review of the status of the debate on the relationship between ethnography and design, I make the case of medication-management to show how it has been performed differently throughout a research and development project in healthcare; as ethnographic practices, as participatory design practices, and finally as actualized simultaneously in interdisciplinary practices.

ETHNOGRAPHY-DESIGN RELATIONS

The use of ethnography in systems design can be grouped into approaches that a) employ ethnography strategically as a way to inform design *directly* and into approaches that b) inform design *indirectly* by developing conceptual and theoretical foundations in CSCW [e.g. 36, 37].

Indirect Relation

The makings of theory and analytical concepts have been central to the development of CSCW. This line of work has its roots in workplace studies where a commitment to analysis as a component of ethnography is prevalent [29, 37]. These studies are typically naturalistic, ethnographic studies, of complex cooperative environments, where people are considered inseparable from the situated and contingent unfolding of activities in which tools and technologies are used. Generally, it is agreed that workplace

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ethnographies and the making of analytical concepts are “critical” and they have had, and continue to have, a profound impact on CSCW systems design [15, 33, 37]. However, this is not in the form of a direct relationship such as requirement specifications but in indirect and more diffuse forms, which makes it difficult to demonstrate explicitly [36]. For example Suchman’s influential study and coining of ‘situated action’ have enlightened and attuned designers in many ways [ibid.]. However, the histories of the trails of this impact remain mostly implicit in papers reporting on findings [37]. The principal motivation behind the analytical and conceptual development in CSCW is therefore the long-term contribution to the reshaping of how we conceive everyday social actions and interactions in the workplace [29, 37] – and not the direct impact this may have on systems design.

Direct Relation

However, ethnography has also been employed much more strategically as a way to mediate directly between users and designers. In CSCW and HCI, the term ‘ethnography’ has become shorthand for fieldwork investigations that to some extent are in situ, qualitative, or open-ended. It includes a wide range of specialized methods, such as Contextual Inquiry [24], which most often do not aspire to following the principles of what is considered traditional ethnography.

Proponents of more traditional versions of ethnography have been critical of this “misconception” of the term and have argued for the value of keeping to the classic approaches and principles [5, 14, 15, 18]. Ethnomethodologists in particular [e.g. 10, 35] have claimed that ethnography can provide access to users’ needs and make visible the relevant action and interaction of a workplace through detailed analyses. Here, ethnography is typically operated in a hierarchical mode as a *subordination-service* [6] to design where analytical insights are made available through a process of translation, thus maintaining the principled ethnographic approach. Approaches to bridge the gap [33] between users’ context and designers’ activities include iterative incorporation of fieldwork analysis into a “requirements capture process” [35] and securing ways of effective communication: “*Our purpose is to inform systems designers – i.e., those parties who are actively involved in the development of computing systems and applications [...]*” [14].

The most common form of translating analytical points to inform design is through the development of design principles, rules, or guidelines [36]. In CSCW and HCI, this approach is widespread and is particularly visible in papers in which a section on ‘implications for design’ is included. However, as workplace ethnographies (with no ambition of informing design directly) are considered “critical”, approaches that employ ethnography to mediate directly between users and designers are considered “problematic” by many [5, 9, 15, 35-37].

Anderson [5], for example, calls it a misunderstanding to use ethnography as a method for accessing information about users and as a method to capture requirements. He reminds us that “[w]ithin the social sciences, ethnography is a form of reportage and not a form of data collection” and “[a]t bottom, ethnography is a literacy practice: an analytic strategy that deploys certain modes of representation.” He argues that the failure to appreciate ethnography’s analytic aspirations misconstrues the character of ethnography and consistently subverts the analytic component of ethnography. Dourish [15] revisits Anderson’s argument in a critique of the approach of “implications for design” and argues that translating analytical insights to design implications does not succeed in utilizing the potential of ethnography in systems design: “... a bullet list of design implications formulated by an ethnographer is not the most effective or appropriate method” [ibid.]. The sympathy for the merits of traditional ethnography and the critical position towards the ‘misuse’ of the social scientific heritage is also visible in the labeling of remodeled fieldwork methods such as *quick-and-dirty ethnography* [35], *scenic fieldwork* [10], or *quasi-ethnographic work* [18].

THE PERFORMATIVE TURN

Whether traditional ethnography is considered well or ill suited for the purpose at hand – the logic, of which the discussion is turning, is epistemic. What is discussed is essentially whether or not ethnography is a fruitful approach to *mediate between users and designers*. The ethnography-design debate is about “user-designer relations”, as Hartswood et al. [22] calls it and the challenge of designing useful systems is framed as a matter of better understanding the users. It is “endemic”, as Rogers [36] calls it, that social scientists are seen as being in a much better position of understanding and articulating the users’ concerns.

This logic of problematization introduces a set of problems, which in this paper, I propose ways to overcome. The first problem is the construction of a separation between design and use. Focusing on the user-designer relation suggests a design process where design happens away from users and situations of use and where designers, as a consequence of being distant, need some kind of representations of users and their situational context to do their work. It suggests that “*once we get an accurate representation, we will solve [the users’] problem*” [8]. This stands in stark contrast to a process “*where the design practice and the designed-for practice are not separated in time or space.*” [ibid.] Creating useful systems by focusing on creating good representations of users and their problems (such as ethnographic descriptions) has limitations - particularly because it installs a separation of design and use (see [8, 22, 23] for a similar critique).

The second problem is related to the first. It is the conceptualization of the relationship between ethnography

and design as an ontological gap to be bridged. That is, to some researchers, ethnography is basically a literacy practice that does not suit the purpose of design, whereas to others, ethnography is considered useful if just we can find effective ways of translating and bridging the gap between a descriptive and prescriptive practice i.e. *in what ways can representations of ethnography (theory, concepts, description) be useful for design*. For these reasons, the debate has considered the relationship in terms of direct or indirect and that ethnography, at heart, is an epistemic practice that works by creating representational descriptions of the world. It therefore becomes troublesome to mobilize the critical value of ethnography since it, ontologically, is depicted as entirely different from design. Instead of positioning ethnography on a par with design in the work of designing useful systems, the ethnography-design relation is framed as a problem of managing ontological differences. However, attempts to “challenge divisions” [34] have been made and to come “full circle” [7] by performative ethnography and participatory design in situations of use [e.g. 9, 21, 23, 26, 40].

Design Interventions and Co-Realization

In the early studies at Xerox Park Suchman, Blomberg, Orr, Trigg, and others [40] have suggested *design interventions* as a work-oriented and cooperative strategy for novel technology production. Through this disciplinary interworking they propose a research strategy that “*establish relations with specific work settings and to use those as sites for cooperative applications design*” [40]. With a similar ambition, Karasti [26] suggests the projection of design suggestions into the users’ lived experience of the future practice, thus simulating *technology-in-use*. Büscher, Gill, Mogensen, and Shapiro [9] propose a *bricolage approach* as a way to develop the relationship between ethnographic and participatory approaches to design. By empirical examples they propose this as “*developing CSCW ‘in practice’*” through *situated experimentation* and “*careful attunement to the circumstances of work*”.

Hartswood et al. [23] acknowledge the fundamental problem of letting ethnomethodologists act as a “proxy” for users i.e. making visible the detailed aspects of a workplace as an informational input and as part of requirements capturing. They suggest overcoming these troublesome user-designer relations by creating a “shared practice” between users and IT professionals that is grounded in the lived experiences of users. *Co-realization*, they call it.

ONTOLOGICAL RECONCEPTUALIZATION

As a way to explore how the critical value of ethnography in design could be otherwise, I engage in *ontological reconceptualization* [25]. Instead of framing the problem of the ethnography-design relation as being a matter of representing the user, I frame the problem as being a matter of how ethnography and design can be *performed* simultaneously in situations of use. Basically, this is a turn

to away from considering the problem as epistemic. By a performative ontology users’ needs are not considered to be “*out there waiting to be revealed*” [20] or described and re-presented. Rather, users’ needs are continually in the making and *emerge* as effects of interworking ethnography and participatory design.

To make this case I present an approach where we (design researchers) work deliberately to overcome the disciplinary boundary between ethnography and participatory design. In many ways our approach resembles the methods presented above, where a reflexive relationship between ethnographic researching and in-situ appropriation of technologies in situations of use engages in co-realization. However, in other ways I suggest a more radical mobilization of ethnography and design. I want to suggest a shift in thinking and practicing ethnography-design relations as an orientation towards *effecting ontological change* [6]. By this I mean to give up on the ontological differences and to begin making ethnography, design, and users’ practices mutually constitutive as interdisciplinary practices. That is, to start considering participatory design practice as ethnographic practice and vice versa.

To enable an investigation of the ways in which ethnography and design have methodologically converged in our project, I turn to later developments in Science and Technology Studies (STS) where ‘performativity’ has been highly developed [28, 30]. Central to this philosophical shift is moving focus away from depicting ethnography as only producing representations (theory, concepts, descriptions) towards a concern with the actual physical encounter, the practice, and the real time action and activity of ethnography. Law [28] argues that (ethnographic) methods in practice “*not only describe but help to produce the reality that they understand*.” Callon explains this as: “*Scientific theories, models, and statements are not constative; they are performative*” [12]. Turning to performativity, then, also has methodological consequences for design where representations (sketches, models, prototypes) are also no longer considered the ultimate product, but rather as equal actors in the practical making of ethnography, design and use.

This ontological reconceptualization means that ethnography should not be about acquiring knowledge in an epistemological way – as a matter of creating valid descriptions – but rather as a matter of attending to the multiple ways that objects are performed [30]. In her book, *The Body Multiple: Ontology in Medical Practice*, Mol [30] makes the argument that the illness of atherosclerosis is not singular, but multiple. According to Mol, there are *multiple* atherosclerosis i.e. each site produces its own version of atherosclerosis: At home, the patient experiences it as pain in the lower leg; in the radiology department atherosclerosis emerges as a relation between the body, the x-ray machine, and other artifacts; in the doctor’s office atherosclerosis emerges as a relation between the desk, the chairs, the

doctor and the letter from the radiology department. In short, practices produce ‘atherosclerosis’ differently. The performativity and multiplicity of reality has methodological implications for experiments, where ethnography and design are deliberately brought together in situations of use. Mol, Law and others’ work signify a shift in investigating the doing, the being, and the ways in which objects emerge – or said otherwise; how objects are continuously *in the making*. For the project of this paper, I make the case of medication management in the making by describing an experiment of how ethnography and design have been performed in situations of use. To do this I examine the emergence of the concept of ‘medication management’ (as one among many concepts) on a research and development project in healthcare by asking the question: *How is medication management performed differently, where, with what tools, and by what practices?* By investigating how medication management emerges relationally between practices of design, ethnography, patients, and clinicians, it is made possible to illustrate how methods in later phases of the project have converged.

The analysis is divided into three figurative stages to foreground the transformation of methods and techniques from traditionally different disciplines and to show the consequences of their convergence. In the first stage I highlight how the concept of ‘medication management’ emerges as an object of ethnography in a more traditional sense. In the second stage I do the same and look at how practices of participatory design perform medication management. In the last stage, I illustrate how it gets performed as ontological effects between ethnography, participatory design, and patients’ and clinicians’ practices.

‘MEDICATION MANAGEMENT’ IN THE MAKING

Co-constructing IT and Healthcare (CITH) is a combined CSCW and participatory design project in healthcare. CITH has engaged distributed healthcare actors in treatment of heart failure patients with an ICD (advanced pacemaker) in the Copenhagen Region, Denmark. The first period of the project, 2008-2011, has involved an interdisciplinary project group consisting of 11 researchers with backgrounds in e.g. Cardiology, Computer Science, and Sociology. The sub-project I report on here [see 31 for the other sub-project] involved 4 CSCW/participatory design researchers, more than 50 patients and their relatives, as well as a heterogeneous group of health professionals; 4 nurses, 4 bioanalysts, 5 Secretaries, and 7 cardiologists. The care for ICD patients is distributed according to specialty; medical care is carried out at local hospitals, whereas ICD device-centered care is carried out at the Heart Centre at the Copenhagen University Hospital. Initially, the project was open-ended with no technological solution imagined but as the project moved through various stages of experimental, explorative and interventionist design research, six consecutive prototypes of a web-based personal health record was developed and used as an actual part of the participants’ health care.

Ethnographic Practices

In the first stage of the project we carried out ethnography in a more traditional sense. This involved fieldwork; *in situ* observations and interviews at the participating hospitals and in patients’ homes. Focus was on getting access to field sites and creating a shared understanding of the distributed treatment organization and practices of heart patients with an ICD. By fieldwork investigation and analytical activities we established insights that proved to be vital for the development of what came to be a web-based prototype of a personal health record. In this work, a relatively mundane fieldwork tool, the patient interview guide, became useful also for project collaboration when preparing for and conducting field visits. The interview guide was most often accompanied by an audio recorder, a notebook, and a camera. An early version of the document shows an interest in patients’ management of medication: *“How is medication managed? Which medicine? What are your experiences from taking the medicine?”*

Cooperative Activities; Sharing Symptom Experiences

Through initial field trips to ICD patient homes, we quickly found that the management of medication is an innate part of their everyday practices. Consider an extract from an early field trip to Hans’ home (an invented name of a participating patient. All names in quotes and stories are anonymized).

Well into the interview, the Design researcher asks Hans: “Is there any medication you take every day?” Hans: “Oh, yes. I have to take 14 pills a day. Something called Dimitone [heart medicine; beta blocker], I have to raise the dosage every day – right. I need to reach 25 mg twice a day, but I have a hard time doing it, because I get so dizzy.” Design researcher: “How do you get a hold of the medicine – do you need a prescription?” Hans: “Well, I call my own doctor and he rings the pharmacy.” Design researcher: “And then you go to the pharmacy?” Hans: “Oh no, then my friend fetches it. Or... yes, sometimes I take the bus down there. Completely helpless, I’m not yet.” (Transcription from audio recording, November 2008)

From this interview and others, we learned that the majority of the patients take a variety of pills at different times every day. The individual act of pouring a glass of water and swallowing the pills is just a minor part managing medication. The work of keeping track of how many pills are left and getting the medication home is a collaborative activity. Hans needs to contact his GP to prescribe the medicine and sometimes relatives or friends help by going to the pharmacy in opening hours and collecting medicine to Hans.

Medication management is also about careful leveling of dosage. As with Hans, experienced symptoms and side effects have crucial consequences for everyday life such as dealing with side effect of dizziness and being tired. Experienced symptoms and side effects are fundamental to

the practice of medication management: *Can I cope with taking the high dosage of Dimitone today as the doctor has prescribed? Maybe I won't experiment tomorrow when I leave the house to go shopping.* Moreover, such experiences are not only relevant for Hans' and other patients' practices of medication management at home. Experiences become essential information in the cooperation with cardiologists in medical treatment. This is highlighted in the following extract from an early observation of medical treatment – a cardiologist-patient consultation, in a regional hospital of Copenhagen.

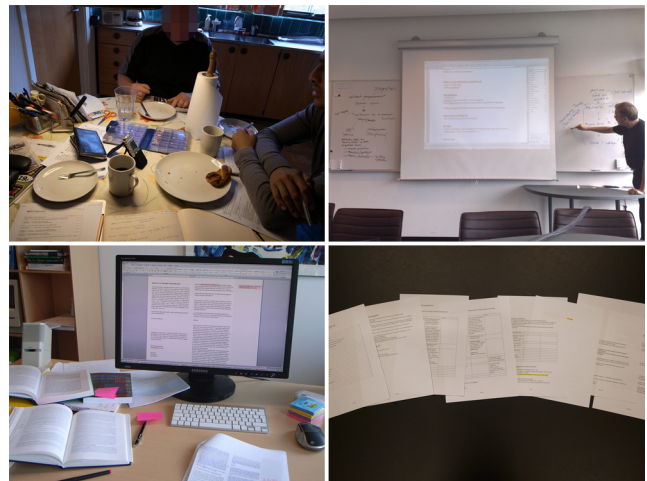
Not long into the conversation, Kurt says: "I'm a bit more worried about my diet situation - I've lost five kilos in 2-2 ½ months. I was so proud when I reached 70 kilos, now I'm down to 65. It's obvious, because the food runs straight through me". The cardiologist says: "Hmm.... do you have any idea which of the pills we've given you makes the problem?" Kurt and the cardiologist discuss back and forth about Kurt's experiences and settle on a drug called Selo-Zok. Then they discuss whether or not Kurt should stick to the high dose. Kurt explains that after the increase in dose he is able to walk to the third floor without any problems. But now he has trouble with his stomach. Nonetheless, the cardiologist insists that he sticks to the high dosage. (Transcription from audio recording, June 2009)

From this extract we see that Kurt has correlated the increase in dose of Selo-Zok with improvements in his physical condition. Living on the third floor is no longer a problem because he feels the symptoms of shortness of breath are disappearing. This is crucial information for the cardiologist since it reaffirms that the medicine is working. However, Kurt expresses his concern about the possible side effects; poor appetite and diarrhea. He indicates that he has kept track of his weight and that he lost five kg. This information, as well as Kurt's request of a prescription, makes the cardiologist write, stamp and sign a prescription for a different but related drug with supposedly fewer side effects. Fundamental to Kurt's successful medication management is therefore to monitor, and importantly, share why, when and where, symptoms and side effects are experienced. Medication management is inherently cooperative and involves good memory that is very often supported by conversations with relatives and writing down (on paper) symptoms, side effects, and questions. Successful medication management, then, is also about successful information management.

Reproducing the User-Designer Model

At this stage, Hans' and Kurt's ways of managing medication are left relatively untouched. The interviews and observations have had very little, perhaps no effect on the way they manage medication. Nonetheless, the field visits had affected our (design researchers') work on the project. To a large extent, we had been doing ethnography as practiced in CSCW; reviewing the literature on patient

work; doing fieldwork observations and interviews; writing naturalistic descriptions and analyses; and documenting design implications. What I have just presented is a compressed version of what we mean by 'traditional' ethnography and according to Mol's [30] philosophy of multiplicity it exemplifies how ethnographic work (using audio recorders, interview guides, writing analytic working papers etc.) has *performed* medication management so far.



Collage 1. Ethnographic Practices [clockwise]: Interview about medication management in patient's home; collaborative analysis; writing up analysis; working papers;

An epistemic mode of practicing ethnography is reproduced and to a large extent it *keeps patients and health professionals and design researchers' practices apart* [For a similar critique, see 17]. The product of research is mostly materialized as text in digital documents and performed at sites away from the ones studied. The ethnography-design relation has so far reproduced the modeling of the problem as a matter of user-designer relations (see collage 1). The ontological differences have been sustained. So far we had employed ethnography as a subordination-service to design [6], using it as a proxy to provide access and insights about 'the user'.

Participatory Design Practices

With similar design interventionist intentions as Suchman et al. [40], Karasti [26], Büscher et al. [9], Hartswood et al. [23], and Halse et al. [20] we turned to Participatory Design and cooperative prototyping to look for methods, tools and techniques that could make the project more change oriented. Essentially, we wanted to not just think, write and discuss medication management, but also to simultaneously design and actualize an improved form of medication management. As we shall see, the project therefore came to include co-design workshops and sessions, web-page sketching and cooperative prototyping of a patient-centered web-application and we began to put more effort into promoting the project (i.e. making graphically inspiring invitations to workshops and making sure participants felt empowered through participating in the project).

Design Game

To begin a transition towards actualizing design and ethnography as part of a use situation, we carried out what we called a situation-card workshop. We arranged a workshop to co-explore and validate insights from the ethnographic activities and to enable co-design of solutions to experienced problems. Inspired by design games methods in Participatory Design, we developed a game board and twenty ‘situation cards’ divided into four themes (e.g. ‘Medication’ and ‘Capturing Symptom Experiences’) with an introductory text and a selection of related questions. Through the design game workshop, it became clear that the patients have medicine prescribed from various institutions and that it is the prescribing institution’s responsibility to follow up on the specific drug. The format also provided for discussions of the idea of having one’s own medicine list written down; for example on the computer or as a homemade list of medicine to fit the wallet (see collage 2). The participating patients agreed that an online patient-centered platform potentially could be a tool to support them in managing medication.



**Collage 2. Participatory Design Practices [clockwise]:
Design game; a patient’s homemade medicine list; co-design
workshop at university; co-design session in patient’s home;
design intervention at the Heart Centre**

Now, medication management had been performed at the university with a cardboard game prompting discussions of the different ways patients keep track of medicine. It could be said that the interview guide from fieldwork activities had translated and materialized into a cardboard game and instead of asking about previous experiences it had been supporting the envisionment of different ways of managing medication. The design game reproduces more future-oriented tradition of Participatory Design [39].

Prototyping: Towards Design-in-Use

As part of prototyping and having interventionist ambitions, we decided that we needed to begin prototyping if we were to get answers on the real value of the ideas that were emerging. We were interested in working towards the integration of ethnography and design in situations of use

i.e. *design-in-use* [16, 23]. We commenced an iterative development approach and made the first version of the web-based personal health record, the prototype, work as a combination of a tool for doing design research and a diary-like tool for patients to keep track of symptom experiences. The first version was a relatively simple web-application built on an open-source content management system. It had three features where patients could make different kind of text entries. The first prototype was therefore flexible in terms of the functionality it stipulated, leaving room for patients’ own interpretative of use. Within four months we had employed a PHP web developer and sketched, implemented, and held two launch workshops at which we provided twelve patients and relatives with a login to their own profile in the prototype. We asked them to use it to write about symptom experiences and to prepare for medical consultations by e.g. noting down questions.

The twelve patients who signed up to use the first version wrote various things, including experienced symptoms, questions in relation to their ICD, summaries of consultations, and other diary entries. The majority wrote about their medication and one patient entered his own complete version of a medicine list. The list reveals how a participant (or ‘user’) performs medication management by explicating at what time a number of pills should be taken (morning, evening), their intended effect (e.g. diuretic, rhythm stabilizing, anticoagulant), how many milligrams, and the name of the generic drug name (e.g. Cordan):

“Morning Evening For:

*Selo-Zok (metopolosuccinat) 1A Pharma 50mg 1
rhythm-stabilizing*

Spiron 25mg 1 diuretic

Trandolapril 0.5 mg 2 strengthens the heart

Aspirin 75mg 1 against thrombosis

Furix 40mg 1 diuretic

Kaleorid 1 750mg potassium supplements

Cordarone (Cordan) 200mg 1 rhythm-stabilizing

Simvastatin 40mg 1 cholesterol-lowering

Marevan 2.5 mg Agreement anticoagulant”*

(Transcription from a patient’s writing in the Experience Diary feature of the first prototype [translated from Danish])

As a way to qualify what the participating patients had written, we arranged for co-design sessions in patients’ homes. In this way, we could combine elicitation studies (discussing patient-generated information) with co-designing features of the second version of the prototype. The result of these co-design sessions was a collective decision to include a Medication Profile feature in the second version of the prototype. To aid sketching the Medication Profile web-page (and two other web-pages), we arranged another co-design workshop with patients at the university, as well as several co-design workshops with cardiologists and bioanalysts from the participating hospitals (see collage 2). After debriefing sessions, mini-

workshops, and co-sketching of web-page features, the web-developer would take time to sit, either at home or in his office on the fifth floor at the university, to program the second version of the web-based prototype. Soon after, the second version of the prototype was launched in collaboration with 23 patients – now with the possibility to communicate with clinicians using a web-module developed for clinicians to view selected patient profiles.

At this stage, the project had moved from focus on creating an understanding of medication management (among other concepts) towards the participatory design of it. A shift in research mode had engaged a range of activities that are different from those of ethnography. Nonetheless, these types of activities are part the participatory design methods-tools-and-techniques catalogue and are in this respect not very exotic. But, in an STS-framing where activities and tools are foregrounded, it becomes apparent that some of the activities had provided for situations where ethnographic practices and the practices of design started to converge. Particularly though patients' use of the prototype and in co-design sessions in patients' homes where it started to become difficult to separate ethnographic inquiry from participatory design as moments of *design-in-use* [23]. The methods had started to converge in situations of use.

Interdisciplinary Practices

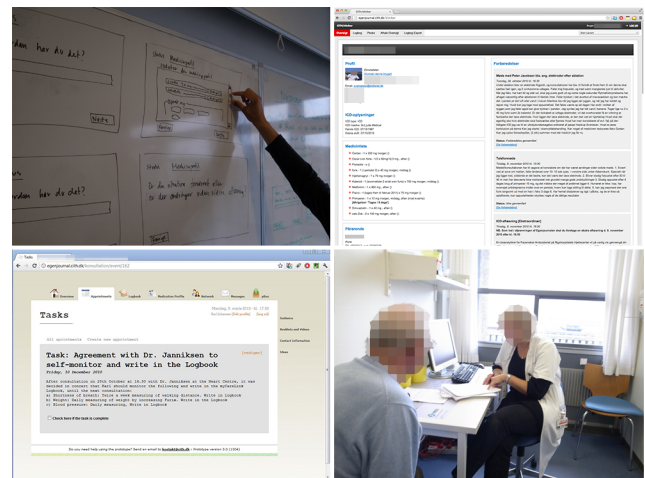
The second version of the prototype included features that made it possible to trial it as a collaborative tool and as an integrated part of health care. In this way, the ethnographic studies of the prototype in situations of use changed radically because the boundaries between ethnography, design, and use were severely blurred. It was no longer fruitful to characterize method activities as belonging to one or the other discipline. By employing *Design interventions* [4], we succeeded to create moments where multiple practices were performed simultaneously (see collage 3). Situations of trialing the prototype as e.g. part of a medical consultation, were not just moments where we could apply our ethnographic curiosity and proactively experiment with situations of use i.e. inquiring into other ways of performing medication management with the prototype. Design interventions were also situations where the participants and we could intervene and cooperatively re-design or suggest new features in the prototype etc.

By critically analyzing the design interventions with the working prototype, we realized that we had introduced a new category of invisible patient work [3]. With the prototype installed in use, the concept of 'medication management' was expanded, introducing expectations of active patient participation [2]. Patients now needed to log in to the prototype from home and write about their experienced symptoms, side effects and medicine list. They were asked to approve the medication profile they had entered, as a way to inform clinicians remotely. The integration of an analytical dimension (ethnographic reflexivity) in design interventions led to the discovery of

new ways of relating ethnography and design. The prototype had become useful, not only in performing a newly designed and possibly better way of managing medication (e.g. letting patients easily share symptom experiences and side effects with cardiologists through the prototype and improving their overview by a keeping self-maintained medication list in the prototype). The prototype had also become a 'research device' [13] capable of making critical questions performative and thus operationalizing ethnographic inquiry in quite different ways than the interview guide and audio recorder. In Law and Callons' [12, 28] philosophy, the participants and we could create what we want to study. We could engage the ethnography-design relation in other ways than direct or indirect and instead work towards a convergence of the methods. The prototype features installed in use could support *proactive analysis* [41] of medication management *in the making* [11]. That is, we could explore and experiment with emergent forms of medication management instead of working to make a descriptive, retrospective analysis of (current forms of) medication management. Our approach provided for ways in which analysis could be proactively performed on par with participatory design [see also 27].

Prospective Analysis

To experiment with proactive analysis, we sketched and implemented a feature in the third version of the prototype called Tasks.



Collage 3. Interdisciplinary Practices [clockwise]: Sketching the Medication Profile feature; prototype module for clinicians; prototype of the personal health record's Tasks feature; design intervention at the Heart Centre;

It was designed to support collaboration in-between meetings by enabling clinicians to give patients tasks in their personal health record (the prototype). This could be keeping track of side effects or writing down symptom experiences for the cardiologist to read when in need of the information. With the new feature, we could try out possibly better ways of collaboratively managing medication. With Tasks we could also proactively perform ethnographic concepts such as '*diagnostic agency*',

'invisible work' [32] and 'homework' [19], which accentuates implications of increased patient responsibility (see [1] for related use in CSCW). In the following extract, a third version of the prototype was launched and a patient, Karl, agreed to prepare for an upcoming medical consultation. He used the prototype to communicate his medication profile, symptom experiences, and side effects, and in turn, the cardiologist used the prototype and Karl's writing as support for his own preparation. However, neither the cardiologist nor the patient had been introduced to the new feature, Tasks, prior to the design intervention.

The design researcher arrives at the outpatient clinic with the patient and his wife and agrees with the cardiologist to follow him to his office to collect various documents and prepare for the medical consultation. It is the first time they have met, so the design researcher hands over a printed version of Karl's preparation and explains in very brief terms what the project is about and reads the first couple of lines from Karl's summary (5 lines in total). The cardiologist mumbles and reads through the rest of the notes and Karl's medication profile. Then, he collects various documents and test results and call in Karl and his wife to start the consultation. The cardiologist says: "You have completed this [printout from the prototype], and it was actually very helpful because you've explained very well what we really need to talk about ... precisely as you have written three weeks have now past and we have to find out what the plan should be by you telling how you feel". Karl replies: "I've been really ill... air ... and sadness. I've not been able to pull myself together".

They talk through the various symptoms and questions using the prototype printout and at one point Karl's wife interrupts: "But Karl, if I may say something, it's also been worse now that your stomach is bigger". Karl says: "Well, see it's a bit complicated. Now, down here [looking down and pointing to his stomach], I was usually fit and I could see my penis and all the funny parts but dammit, now I can't any more. It means that I have either air or water in my stomach". The cardiologist agrees. Karl continues: "Look, something like that also affects the air in the stomach [pointing at the prototype medication profile]" They are both sitting bent over, looking at the printout. The cardiologist: "Yes it does. You get out of breath more, your appetite gets poorer and so on and so on - if you've got water in the stomach. And the things you've mentioned here... [pointing at the prototype printout]... I think what we do is: We wait and see how things are going in a couple of weeks. You have to exercise and use your body, we increase Furix [the diuretic medicine] because it forces the water out of your stomach ..." [the design researcher takes note of this for use shortly after]

They continue the discussion back and forth about how much they should increase the diuretic medicine, Furix.

After that they spend quite some time talking about side effects and whether Karl should try to decrease another drug; Cordan. They consult the medication profile in the prototype printout several times for the amount of quantities taken. (Transcription from audio recording, October 2010)

Now, the part of medication management, which involves meeting clinicians and discussing symptoms and side effects, has been reconfigured. First of all, the cardiologist encourages Karl to be an active participant i.e. he values getting Karl's notes and medication profile before the meeting. In addition, a new actor – the prototype – has found its place, supporting the discussion around medication. They both use the prototype several times to guide the conversation and get reassurance about certain things. It is not only Karl's explanation that counts; it is also the prototype and what is printed on it. What is interesting in terms of the ethnography-design relation, is that when Karl or the cardiologist point at the prototype, talk about it, or use it to guide the conversation, they at once mobilize earlier ethnographic practices (the interview guide, the working papers, the implications for design where 'medication management' was made 'visible') and participatory design practices (the cardboard game, the home-made laminated medicine card with a medicine list, the patient's written medicine list from using the first prototype and the web developer's PHP code). However, in the situation, these two disciplines come together. One is not without the other. In the design intervention, medication management is different than it was without the prototype.

Performing Ethnography-Design Relations in Use

So far so good. Now, on to the engagement of prospective analysis and participatory design as it happened in the design intervention. At this point in time, the Tasks feature and the related ethnographic discourse on patient work has not been connected to medication management in any situation of use. 'Homework' is still (only) an analytical category that has not yet been realized in situations of use. Then, after almost an hour, the consultation comes to an end and the cardiologist stands up while talking.

The design researcher sees his chance and speaks for the first time: "There is one more thing. We've created something we call 'Tasks', which are a little experimental, but in your discussion with Karl you indirectly ask him to do some tasks – that he must keep an eye on this and that – so, what should he enter in the prototype for your next meeting?" The cardiologist quickly gets the idea and immediately asks Karl to write down (in the prototype), everyday if possible, how far he can walk. This would be helpful in relation to his breathing. The design researcher: "And then there was something in relation to the medication – something with Furix?" and the cardiologist says: "Yes, the weight, walking distance, and blood pressure. Clearly it

would be a great help to have your experiences on that next time we meet”.

They end the conversation and shake hands, and Karl and his wife leave. The cardiologist sits down and says he just needs to finish off and dictate a summary for the medical record. He picks up the tape recorder and dictates: “Memorandum to Karl.... Meets for medical consultation. Full stop. See the above. New line. Since the last meeting. Colon. [the cardiologist looks at the prototype printout] Complaints of oppression, unpleasant upper abdomen [almost exactly as the first line of Karl’s writing]... We try RECIPE CRESENDO [prescription, gradually increase] tablet FURIX to 40 mg x 2 plus KALEROID 40 mg x 2. Daily weight measurement and assessment of walking distance a few times a week. Full stop. Signed [cardiologist name]”.

In these last minutes, the design researcher intervenes and connects Karl’s and the cardiologist’s discussion around Furix (bloated stomach) to an experiment of participatory design and performative ethnography: *Could Karl and the cardiologist possibly want to collaborate in new ways? Do they like this feature at all? What are the consequences of increasingly extending patient responsibility and active participation? How could homework make sense for both patients and cardiologists?* By the design interventionist move of suggesting using the prototype to collaboratively monitor the increase in Furix, the toolbox of participatory design is performed and in one and the same moment, proactive analysis is performed. It goes a bit further because what happens is that the cardiologist agrees with Karl to try to monitor his weight, since it is highly related to the increase in the dose of Furix. And as the cardiologist says: *“Clearly it would be a great help to have”.* The *“Daily weight measurement”*-agreement or the feature Tasks moreover materializes in the hospital’s medical health record through the cardiologist dictating it. And later, which is out of the scope of this paper, it materializes in the prototype by Karl’s self-monitoring and prototype use. By design interventions with the prototype, the ethnography-design relation is blurred and a transcended, proactive way of interdisciplinary practice has taken place. The original problem of modeling the ethnography-design relation as a matter of mediating between users and designers (directly or indirectly) is shifted to become a matter of engaging performative ethnography and participatory design, simultaneously, in situations of use.

CONCLUSION

The methodological problem, which is the point of departure for this paper, concerns the much-debated relation between ethnography and design. I frame the problem as failing to take an interest in use and as considering the ethnography-design relation as epistemic. That is, debating in terms of a direct and indirect relationship and modeling the problem as a matter of bridging the gap between users and designers. Some argue for employing ethnography as

traditionally practiced [e.g. 5, 14], thus creating ontological differences, while others suggest to engage in co-realization [23] and employ design interventions [40] to make reflexive relations between understanding and designing. We have gone similar ways by critically integrating performative ethnography and participatory design as part of situations of use in healthcare. Inspired by later developments in STS I have engaged an ontological reconceptualization and shown how ethnography and design can become co-constituents together with users’ practices in the collaborative making of ‘medication management’. Using the prototype as a research device enables ethnography to be performed prospectively as participatory design and by installing ethnographic concepts in the prototype, ethnography becomes a way of doing participatory design.

By our empirical case I suggest the fruitfulness of turning to performativity and consider ethnography and design as mutually constitutive. I advocate an ontological shift, as is happening in broader lines of social science [6], and to engage ethnography-design relations in proactive ways, making for *ontological convergence*. Working towards performative ethnography and participatory design in situations of use is a novel direction for exploring Schmidt and Bannon’s [38] programmatic call.

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REFERENCES

1. Aarhus, R., Ballegaard, S. and Riisgaard, T. The eDiary: Bridging Home and Hospital through Healthcare Technology. In *Proc. ECSCW 2009*, Springer (2009).
2. Andersen, T. The participatory patient. In *Proc. PDC 2010*, ACM (2010), 151-154.
3. Andersen, T., Bjørn, P., Kensing, F. and Moll, J. Designing for collaborative interpretation in telemonitoring: Re-introducing patients as diagnostic agents. *International Journal of Medical Informatics*, 80, 8, (2011), e112-e126.
4. Andersen, T., Halse, J & Moll, J. Design Interventions as Multiple Becomings of Healthcare. In *Proc. Nordes 2011*, (2011).
5. Anderson, R. J. Representations and requirements: the value of ethnography in system design. *Human-Computer Interaction*, 9, 3, (1994), 151-182.
6. Barry, A., Born, G. and Weszkalnys, G. Logics of interdisciplinarity. *Economy and Society*, 37, 1, (2008), 20-49.
7. Berg, M. The politics of technology: On bringing social theory into technological design. *Science, technology & human values*, 23, 4, (1998), 456-490.

8. Bowers, J. M. The Janus faces of design: Some critical questions for CSCW. in *Studies in Computer Supported Cooperative Work*. Ed. by J. M. Bowers and S. D. Benford, Elsevier Science Publishers (1991).
9. Büscher, M., Gill, S., Mogensen, P. and Shapiro, D. Landscapes of Practice: Bricolage as a Method for Situated Design. *CSCW*, 10, 1, (2001), 1-28.
10. Button, G. The ethnographic tradition and design. *Design Studies*, 21, 4, (2000), 319-332.
11. Callon, M. Society in the making: the study of technology as a tool for sociological analysis. *The social construction of technological systems: New directions in the sociology and history of technology*, (1987), 83-103.
12. Callon, M. What Does it Mean to Say That Economics Is Performative? in *Do Economics Make Markets? On the Performativity of Economics*. Ed. by D. MacKenzie, F. Muniesa and L. Siu, Princeton University Press (2007).
13. Collins, R. Why the social sciences won't become high-consensus, rapid-discovery science. *Sociological Forum*, 9, 2, (1994), 155-177.
14. Crabtree, A., Rodden, T., Tolmie, P. and Button, G. Ethnography considered harmful. In *Proc. CHI 2009*, ACM (2009), 879-888.
15. Dourish, P. Implications for design. In *Proc. CHI 2006*, ACM (2006), 541-550.
16. Ehn, P. Participation in design things. In *Proc. PDC 2008*, Indiana University (2008), 92-101.
17. Fitzpatrick, G. and Ellingsen, G. A Review of 25 Years of CSCW Research in Healthcare: Contributions, Challenges and Future Agendas. *CSCW*, (2012), 1-57.
18. Forsythe, D. E. "It's Just a Matter of Common Sense": Ethnography as Invisible Work. *CSCW*, 8, 1, (1999), 127-145.
19. Grøn, L., Mattingly, C. and Meinert, L. Kronisk hjemmearbejde - Sociale håb, dilemmaer og konflikter i hjemmearbejdsnarrativer i Uganda, Danmark og USA. *Tidsskrift for Forskning i Sygdom og Samfund [Journal for research on illness and society]*, 9, (2008).
20. Halse, J., Brandt, E., Clark, B. and Binder, T. *Rehearsing the future*. The Danish Design School Press (2010).
21. Halse, J. and Clark, B. Design rituals and performative ethnography. In *Proc. EPIC 2008*, (2008), 128-145.
22. Hartwood, M., Procter, R., Rouchy, P., Rouncefield, M., Slack, R. and Voss, A. Working IT out in medical practice: IT systems design and development as co-realisation. *Methods of Information in Medicine*, 42, 4, (2003), 392-397.
23. Hartwood, M., Procter, R., Slack, R., Voß, A., Büscher, M., Rouncefield, M. and Rouchy, P. Co-realization: Toward a principled synthesis of ethnomethodology and participatory design. in *Resources, Co-Evolution and artifacts: Theory in CSCW*. Ed. by M. S. Ackerman, C. A. Halverson, T. Erickson and W. A. Kellogg, Springer London, (2008).
24. Holtzblatt, K. and Jones, S. Contextual inquiry: A participatory technique for system design. in *Participatory design: Principles and practice*. Ed. by D. Schuler and A. Namioka, (1993).
25. Jensen, C. B. CSCW design reconceptualised through science studies. *AI & Society*, 15, 3, (2001), 200-215.
26. Karasti, H. Bridging Work Practice and System Design: Integrating Systemic Analysis, Appreciative Intervention and Practitioner Participation. *CSCW*, 10, 2, (2001), 211-246.
27. Kimbell, L. Closing Keynote Address - Reassembling the visual. *EPIC '08*, 2008, 1, (2008), 316-324.
28. Law, J. *After method: Mess in social science research*. Psychology Press (2004).
29. Luff, P., Hindmarsh, J. and Heath, C. *Workplace studies: Recovering work practice and informing system design*. Cambridge Univ Press, (2000).
30. Mol, A. *The body multiple: Ontology in medical practice*. DUKE (2010).
31. Mønsted, T., Reddy, M. C. and Bansler, J. P. The Use of Narratives in Medical Work: A Field Study of Physician-Patient Consultations. In *Proc. ECSCW 2011*, Springer (2011), 81-100.
32. Oudshoorn, N. Diagnosis at a distance: the invisible work of patients and healthcare professionals in cardiac telemonitoring technology. *Sociology of Health & Illness*, 30, 2, (2008), 272-288.
33. Plowman, L., Rogers, Y. and Ramage, M. What are workplace studies for? In *Proc. ECSCW 1995*, Kluwer Academic Publishers (1995), 309-324.
34. Pors, J. K., Henriksen, D., Winthereik, B. R. and Berg, M. Challenging divisions. *SJIS*, 14, 2, (2002), 3-7.
35. Randall, D., Harper, R. and Rouncefield, M. *Fieldwork for design: theory and practice*. Springer-Verlag New York Inc (2007).
36. Rogers, Y. Reconfiguring the social scientist: Shifting from telling designers what to do to getting more involved. in *Social Science, Technical Systems, and Cooperative Work - Beyond the Great Divide*. Ed. by G. C. Bowker, S. L. Star, W. Turner and L. Gasser, Lawrence Erlbaum Associates Publishers (1997).
37. Schmidt, K. The critical role of workplace studies in CSCW. in *Workplace studies: recovering work practice and informing system design*. Ed. by P. Luff, J. Hindmarsh and C. Heath, Cambridge Univ Pr (2000).
38. Schmidt, K. and Bannon, L. Taking CSCW seriously. *CSCW*, 1, 1, (1992), 7-40.
39. Simonsen, J. and Robertson, T. *Routledge International Handbook of Participatory Design*. Routledge, (2012).
40. Suchman, L., Blomberg, J., Orr, J. and Trigg, R. Reconstructing technologies as social practice. *The American Behavioral Scientist*, 43, 3, (1998).
41. Vikkelsø, S. Description as Intervention: Engagement and Resistance in Actor-Network Analyses. *Science as Culture*, 16, 3, (2007), 297-309.